Basic Principles of Medicine 1
Module: Foundations to Medicine
Lecture No: 4

Tonicity and Osmolarity

Presenter: Dr Janine Paul
Created by SGU Faculty

Learning Objectives

SOM.MK.II.BPM1.1.FTM.3.PHY.0007 Define Fick's Law of diffusion and explain how changes in concentration gradient, surface area, time and distance will influence the diffusional movement of a compound.

SOM.MK.II.BPM1.1.FTM.3.PHYS.0017 Using a cell membrane as an example, define reflection coefficient and partition coefficient, and explain how the relative permeability of a cell to water and solutes will generate an osmotic pressure. Contrast the osmotic pressure generated across a cell membrane by a solution of particles that freely cross the membrane with that of a solution with the same osmolality, but particles that cannot cross the cell membrane.

SOM.MK.II.BPM1.1. FTM.3.PHYS.0020 Contrast the following units used to describe concentration: mM, mEq/l, mg/dl, mg%.

SOM.MK.II.BPM1.1.FTM.3.PHYS.0021 List the typical values and normal range for plasma osmolality. Differentiate between the terms osmolarity and osmolality.

SOM.MK. II.BPM1.1.FTM.3.PHYS.0022 Describe the role of water channels

(aquaporins) in facilitating the movement of water across biological membranes.

SOM.MK.II. BPM1.1.FTM.3.PHYS.0023 Understand how regulation of concentrations of K^+, Cl^-, and other Na^+ solutes influence cell volume.

SOM.MK.II.BPM1.1.FTM.3.PHYS.0024 Predict the movement of water across a

semi-permeable membrane depending on solute concentrations in both compartments.

Recommended Reading

Medical Physiology: Principles for Clinical Medicine, 6e
Rodney A. Rhoades, David R. Bell, 2023; Chapter 2 & 23.

Chapter 2: Water movement across the membrane
Membrane Transport and Cell Signaling | Medical Physiology: Principles for Clinical Medicine, 6e | Premium Basic Sciences | Health Library (lwwhealthlibrary.com)

Chapter 23: Fluid compartments of the body
Regulation of Fluid and Electrolyte Balance | Medical Physiology: Principles for Clinical Medicine, 6e | Premium Basic Sciences | Health Library (lwwhealthlibrary.com)

Fick's Law

- The rate of diffusion of a substance across a unit area(such as a surface or membrane) is proportional to the concentration gradient.

Water Transport

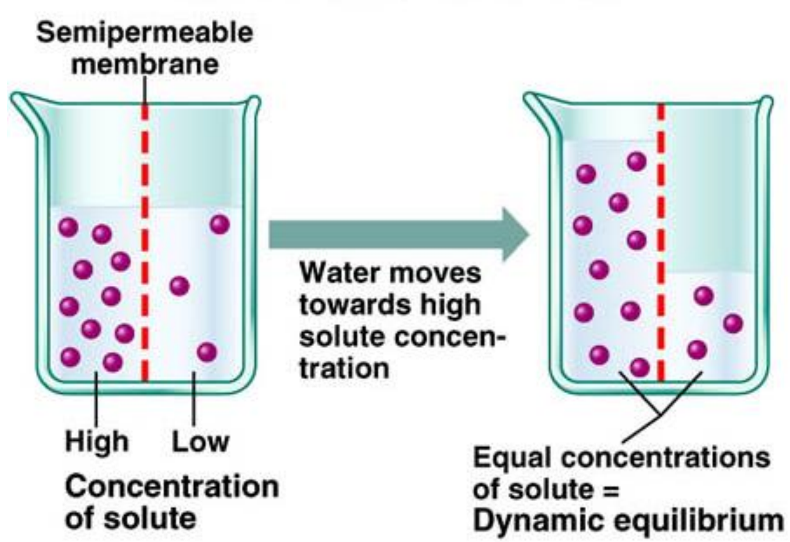
- Entirely passive – no water pumps exist

- Water may pass either through the membrane or via water channels (aquaporins) that exist to allow a greater flux of water across membranes, depending on osmotic gradients

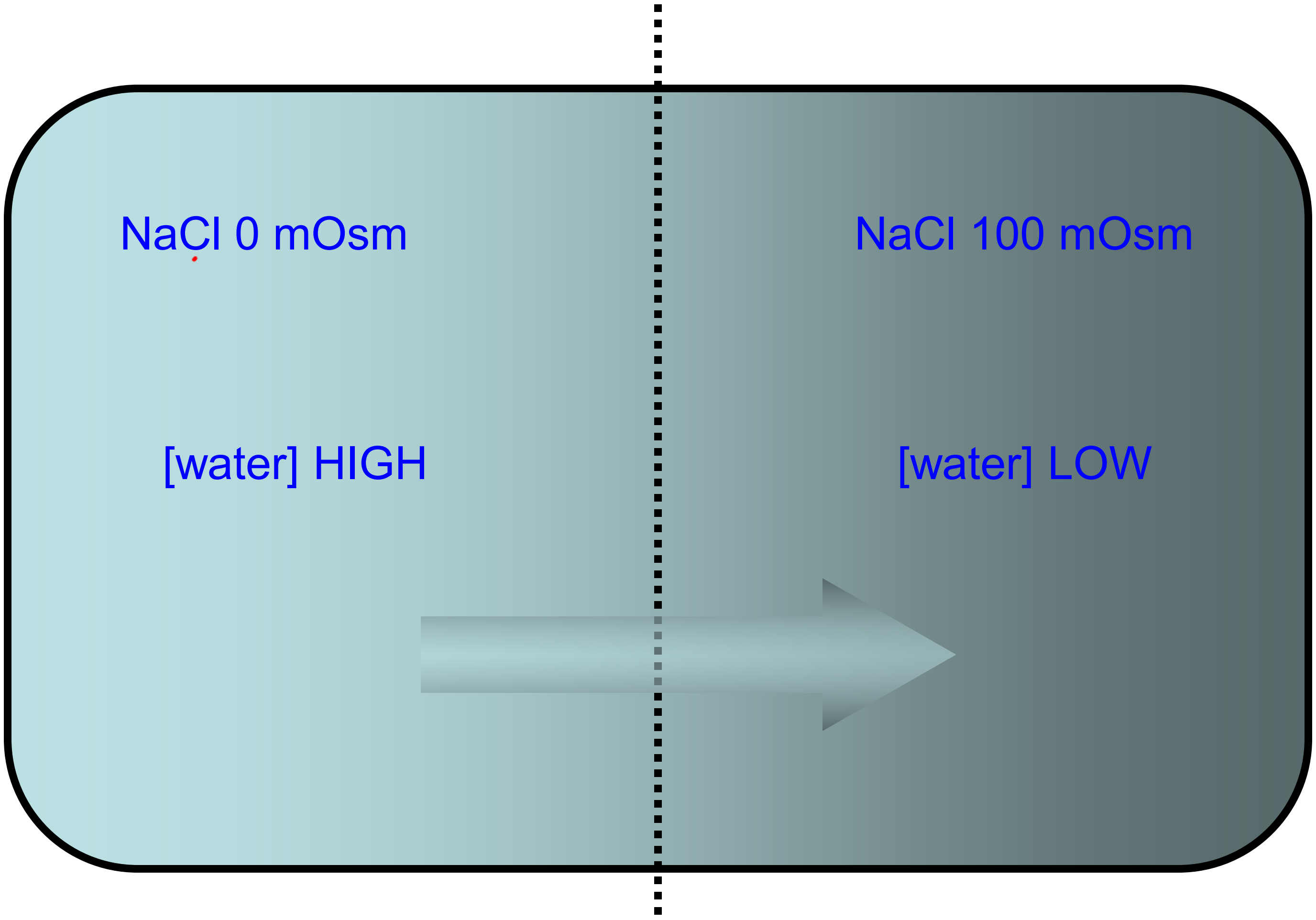
- Because it is simple diffusion, water transport does not show saturation and has linear kinetics

Water Concentration

- Concentration is simply the mass of a substance in a known volume
- Water like any other substrate can have a chemical potential difference – i.e. a concentration gradient between 2 compartments
- Usually we work with the concentration of solutes in water since they are dissolved in water – therefore, the higher the solute concentration the lower the water concentration
- Water too can move across membranes and will do so down its concentration gradient towards area of high solute concentration until it reaches equilibrium



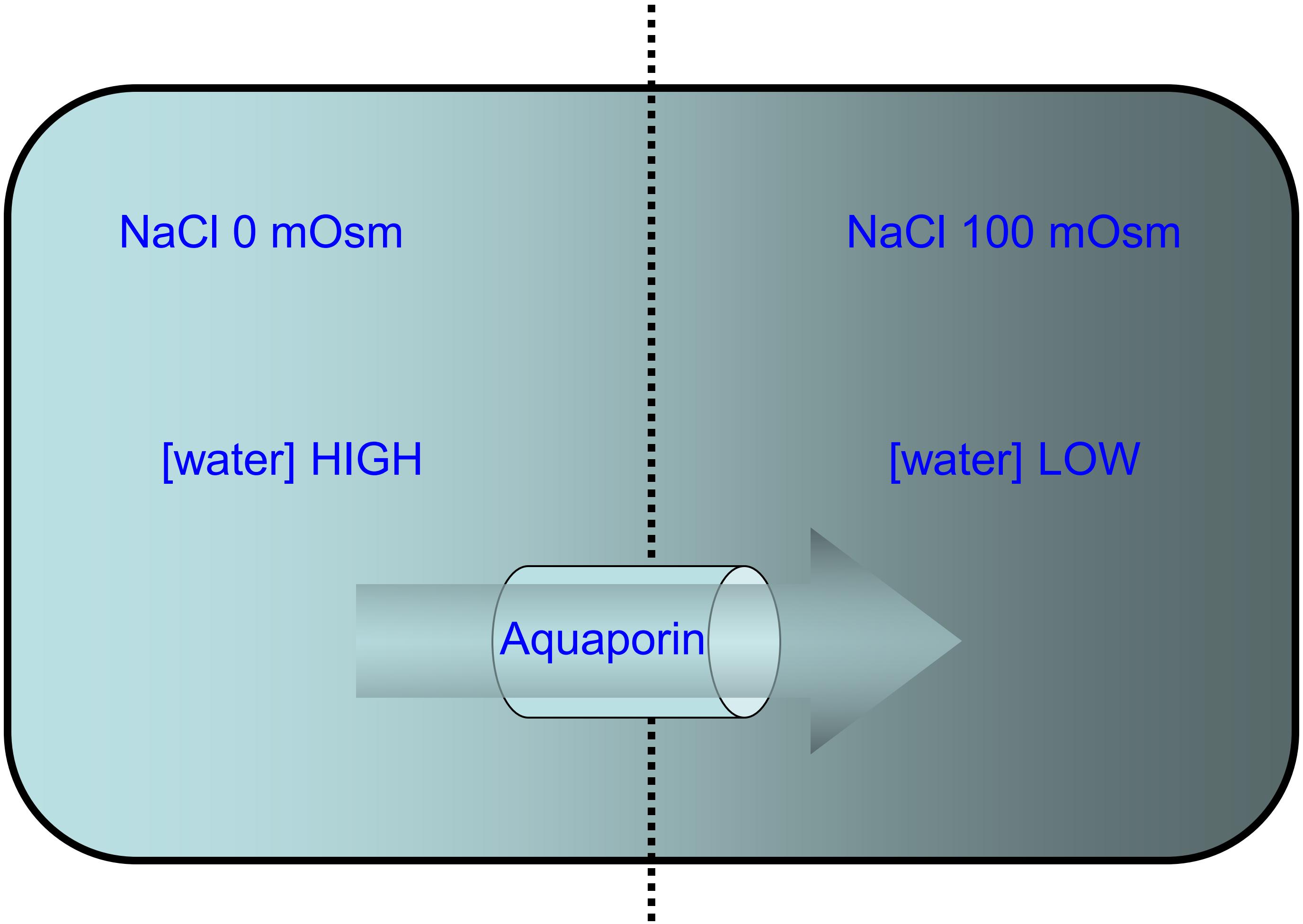
Fluid Transport Across Membranes



If membrane impermeable to NaCl (reflection coefficient (σ)= 1)

Water moves in the direction of low solute concentration to high solute concentration

Fluid Transport Across Membranes



If membrane impermeable to NaCl (reflection coefficient (σ)= 1)

Aquaporins allow passage of water molecules across the membrane increasing net flux of water across the membrane

Driving Force for Water Movement

- 2 major driving forces for water movement within the body:
  - i) Osmotic
  - ii) Hydrostatic driving forces

- The osmotic forces are associated with the concentration of solutes within the fluid -- water wanting to move to an area with high solute concentration

- The hydrostatic pressure is the effect of gravity on the fluid across capillary endothelial cells (CV, Renal)

  Hydrostatic forces across membranes at the cellular level are 0 since membranes cannot withstand large hydrostatic forces water movement is governed by solute concentration→ This is not true of solute transport across vasculature!! Pressures matter!!

Osmosis

- <u>Osmosis</u>: Flow of water between two solutions separated by a semi-permeable membrane caused by a difference in solute concentration → Water will flow from an area of low osmolarity to an area of high osmolarity

- Driving force is a difference in osmotic pressure caused by the presence of a solute and depends on the permeability of the membrane

- Water flux depends upon an osmotic gradient so solute particles that can easily cross the membrane cannot set up a gradient and has no effect upon water flow

- Each substance has a reflection coefficient (σ)

Van't Hoff Equation

- Used to calculate osmotic pressure of a solution

$$\pi = \mathbf{nRTC}$$

π = osmotic pressure (atm)
n = Number of particles produced by a molecule (e.g. NaCl - 2 particles) (mmol/L)
R = Gas constant (0.082 L-atm/mol-K)
T = Temperature (Kelvin)
C = Concentration (mol/L)

The **effective** osmotic pressure (π_{eff}) is modified with σ = reflection coefficient (0-1)

$$\pi_{eff} = n\,\sigma CRT$$

Water Flux

A difference in osmotic pressure between two compartments can provide a driving force for water to flow

$$J_v = K_h A [\sigma (C_i - C_o)]$$

J_v = Volume of water flux
K_h = Hydraulic conductivity (permeability to fluid)
A = Area
σ = Reflection coefficient
C_i = Concentration of total solutes inside the cell
C_o = Concentration of total solutes outside the cell

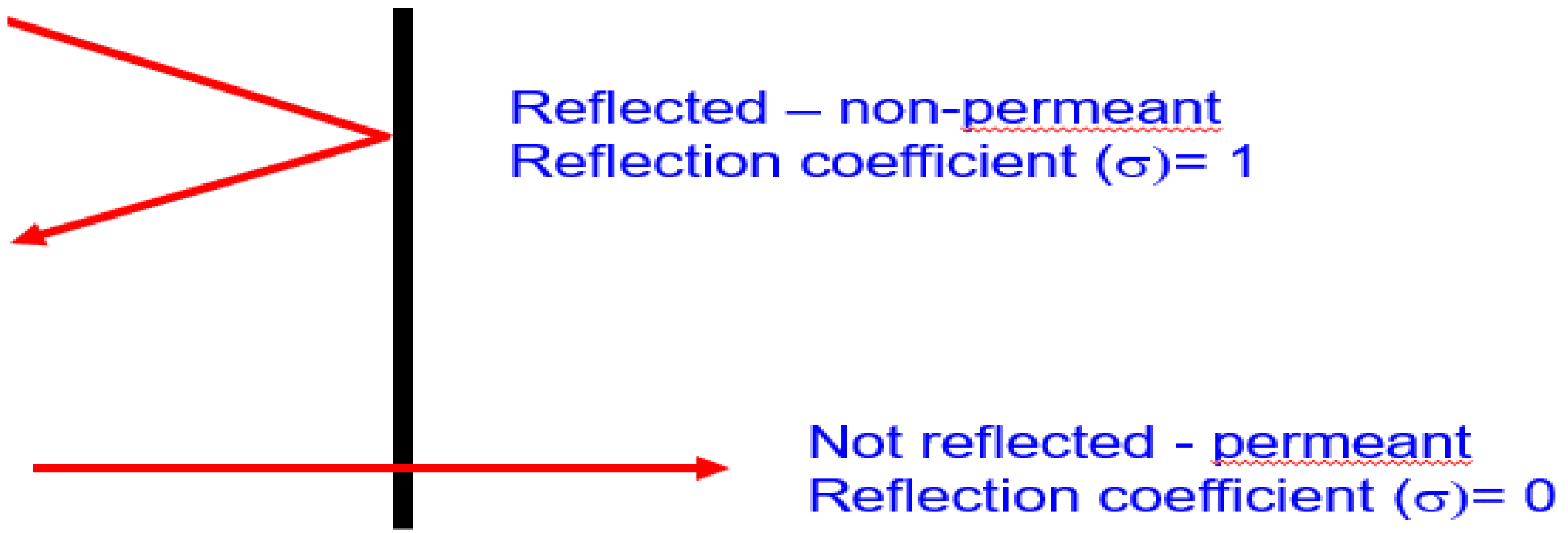
Partition coefficient

- Partition Coefficient = $\frac{\text{Concentration of substance in oil}}{\text{Concentration of substance in water}}$

- A partition coefficient of 1: Substance can exist in both oil and water equally.
- A high partition coefficient >1: Substance can exist in oil (lipophilic or soluble in oil). It can pass easily through the cell membrane .
- A low partition coefficient 1 : Substance can exist in water. It cannot easily pass directly through the membranes .
- Use log values for aesthetics if coefficient is large.

Reflection Coefficient

- Reflection Coefficient (s): How easily a substance can cross the membrane based on its reflection.
- If a substance is reflected by the membrane (does not pass through), then its reflection coefficient is 1
- If the substance is not reflected by the membrane (passes through) then its reflection coefficient is 0 → permeable substances have low (or zero) reflection coefficients



SOM.MK.II.BPM1.1.FTM.3.PHYS.0017
SOM.MK.II.BPM1.1.FTM.3.PHYS.0024

Penetrating vs. Non-penetrating Solutes

- Penetrating Solutes: Can enter cell (glucose, urea, glycerol)
  - Solutes will distribute to equilibrium

- Non-Penetrating Solutes: Cannot enter cell (sucrose, NaCl, KCl)
  - Water will move to dilute solutes

- Determine relative concentration of non-penetrating solutes in solution and in cell to determine tonicity

- It is the non-penetrating solutes that ultimately determine the tonicity of the solution (after the penetrating solutes have equilibrated)

Osmolarity vs. Tonicity

- **Osmolarity**:

Absolute

  - Total concentration of **all** particles in solution

Osmolarity = Φ nC

  - n = number of particles /mol in solution
  - C = concentration
  - Φ osmotic coefficient (usually 1)

Units

1 mM = 1 mOsm

Example

50 mM Na = 50 mOsm Na

- **Tonicity**:

Relative

  - Concentration of only the osmotically active particles
  - Only ***impermeable particles*** contribute to tonicity and cause changes in cell volume
  - Tonicity of a solution describes the volume change of a cell at equilibrium
  - Net water movement will be into the compartment that has the higher concentration of non-penetrating solutes

SOM.MK.II.BPM1.1.FTM.3.PHYS.0021
SOM.MK.II. BPM1.1.FTM.3.PHYS.0023

A solution is:

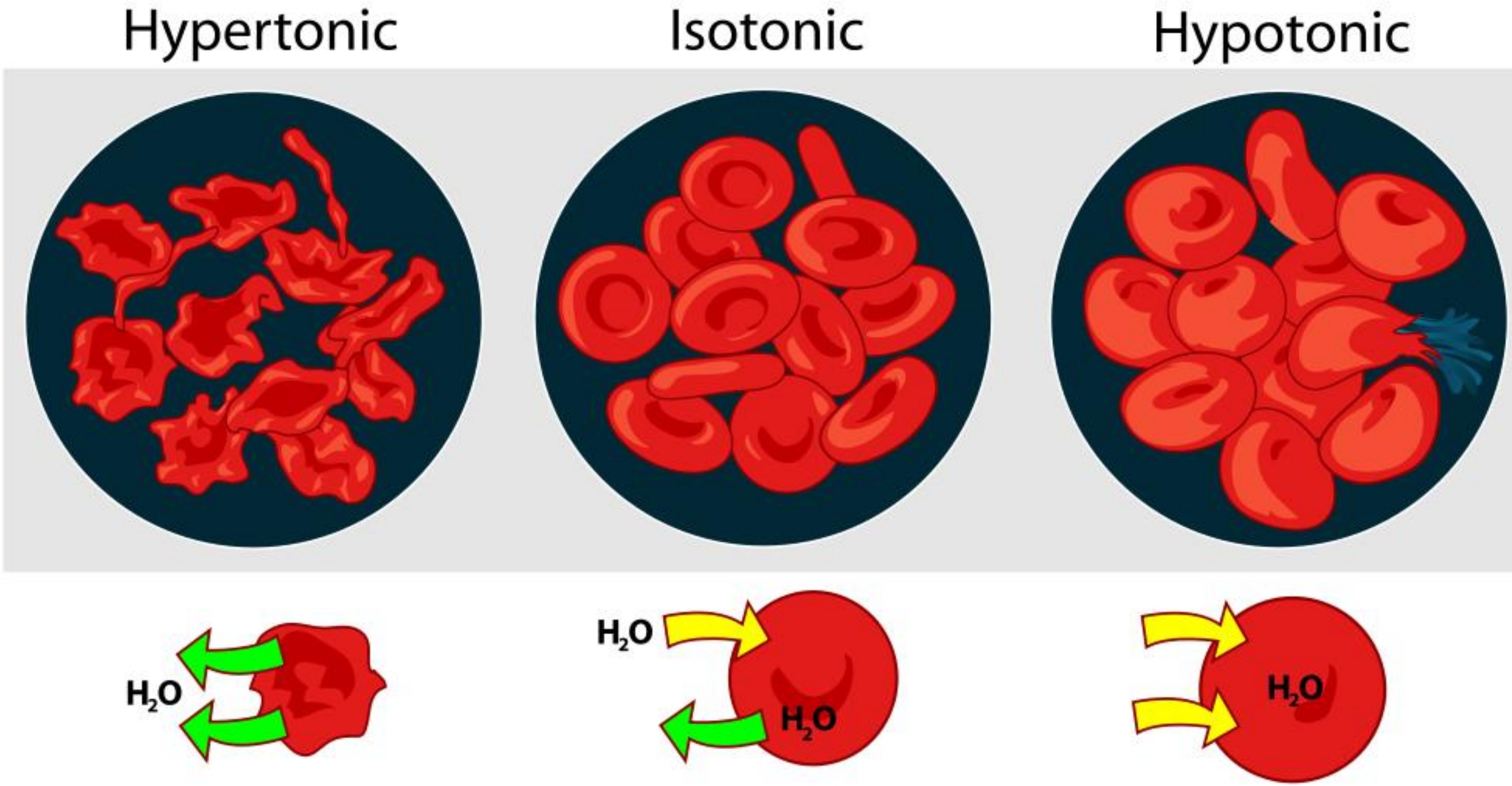
- **Iso-osmotic** if total osmotic pressure of the solution is equal to that of the cell
- **Hyperosmotic** if the solution has greater osmotic pressure than the cell
- **Hypo-osmotic** if the solution has less osmotic pressure than the cell

  - Osmolarity impacts transient changes in cell volume

A solution is:

- **Isotonic** if, at equilibrium, it causes the cell to neither swell or shrink
  - the non-penetrating solute concentrations on both sides of the cell membrane are the same
- **Hypertonic** if it causes the cell to shrink
  - Non-penetrating solute concentration on the outside of the cell is higher than on the inside
- **Hypotonic** if it causes the cell to swell
  - Non-penetrating solute concentration on the outside of the cell is lower than on the inside

  - Tonicity impacts long term steady state of cell volume (after the penetrating solutes have equilibrated)



SOM.MK.II. BPM1.1.FTM.3.PHYS.0023
SOM.MK.II.BPM1.1.FTM.3.PHYS.0024

Solutions

- No volume change – isotonic
- Donate water to cell – hypotonic
- Sequester water from cell – hypertonic

In normal functioning cells the following rules are true…..

1. All hypo-osmotic solutions are hypotonic

2. Iso-osmotic solutions can be hypo or isotonic

3. Hyper-osmotic solutions can be hypo-, iso-, or hyper-tonic depending on the ratio of non-penetrating solutes between the 2 compartments

- Clinical relevance: IV fluid therapy – determining which saline type to use

Intravenous Fluid List

Isotonic Saline Solution

- 0.9% Saline
- 5% Dextrose in 0.225% Saline
- Ringers Solution

Hypertonic Saline Solution

- 10% Dextrose in Water
- 3.0% Saline
- 5% Dextrose in 0.45% Saline

Hypotonic Saline Solution

- 0.45% Saline

SOM.MK.II. BPM1.1.FTM.3.PHYS.0023
SOM.MK.II.BPM1.1.FTM.3.PHYS.0024

Summary

- 2 major driving forces for water movement within the body:
- i) Osmotic
- ii) Hydrostatic driving forces

- <u>**Osmolarity**</u>: Total concentration of **all** particles in solution

- <u>**Tonicity**</u>: Concentration of only the osmotically active particles
  - Only impermeable particles contribute to tonicity and cause changes in cell volume
  - Tonicity of a solution describes the volume change of a cell at equilibrium
  - Net water movement will be into the compartment that has the higher concentration of non-penetrating solutes

- If, at equilibrium the cell volume has not changed – Isotonic solution
- If water is drawn <u>into</u> the cell – Hypotonic solution
- If water is drawn <u>out of</u> the cell – Hypertonic solution

Cell Volume Changes

Example 1: 150 mM NaCl + 50 mM glycerol

- What is the osmolarity?
- What is the tonicity?
- Draw a cell representation of water movement.

<u>Example 2</u>: 150 mM NaCl + 300 mM Urea

What is the osmolarity?

What is the tonicity?

Draw a cell representation of water movement.

Example 3: 300 mM Urea

What is the osmolarity?

What is the tonicity?

Draw a cell representation of water movement.

SOM.MK.II. BPM1.1.FTM.3.PHYS.0023
SOM.MK.II.BPM1.1.FTM.3.PHYS.0024

Example 4: 170 mM NaCl + 10 mM Glycerol

What the is osmolarity?

What the is tonicity?

Draw a cell representation of water movement.

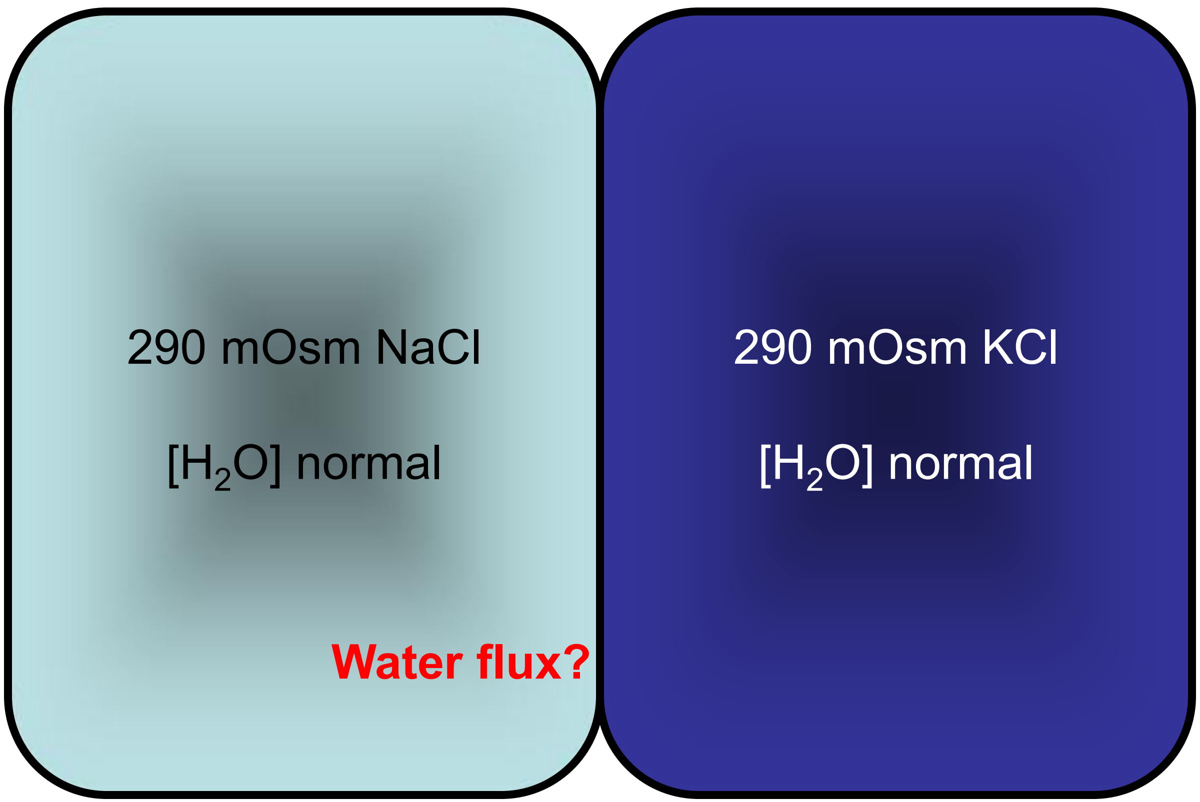
SOM.MK.II. BPM1.1.FTM.3.PHYS.0023
SOM.MK.II.BPM1.1.FTM.3.PHYS.0024

Example 5

Osmolarity/Tonicity

1 L ECF

1 L ICF



Osmolarity?
Tonicity?

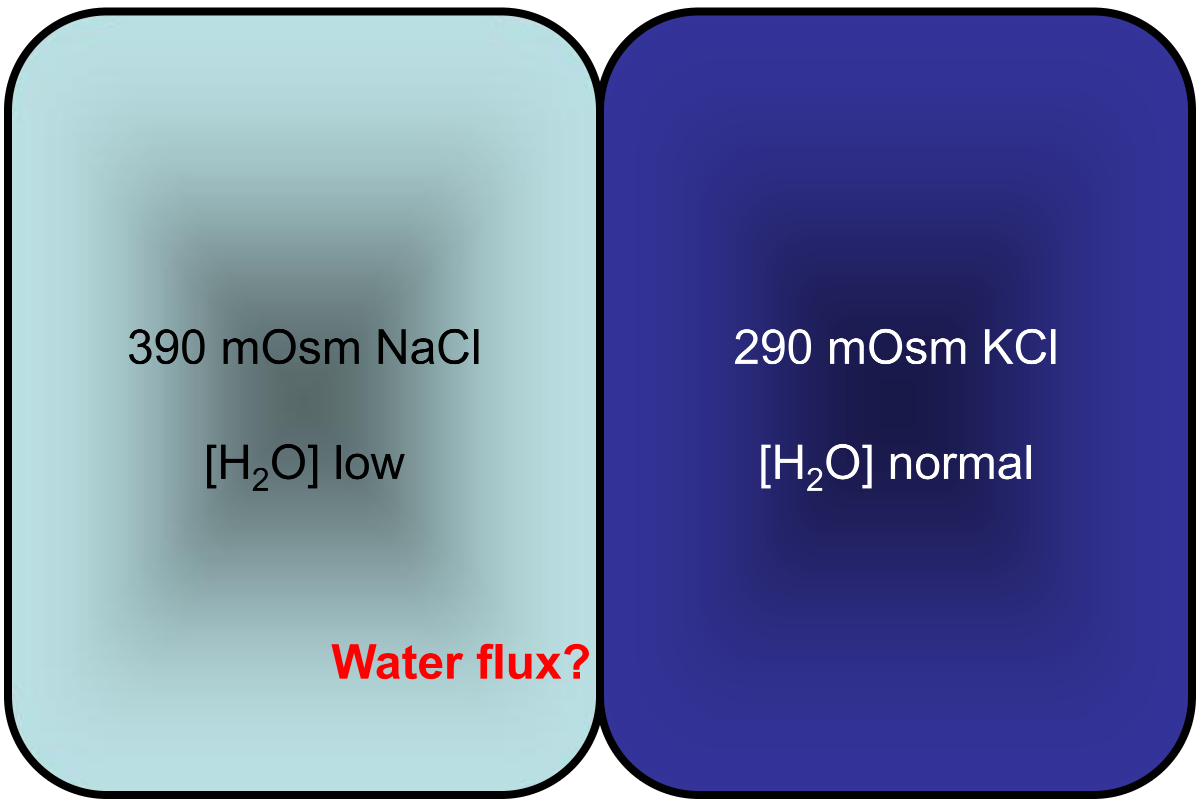
Cell volume After Equilibration?

Example 6

Osmolarity/Tonicity

1 L ECF

1 L ICF



Osmolarity?
Tonicity?

Cell volume After Equilibration?